

Input Manipulation Attacks

CompTIA SecAI+: AI Security Certification Complete Exam Prep 2026 | Section 15: AI Attack Vectors and Evidence Analysis

1. The Core Concept — Why AI Sees Differently

Human perception and AI perception are not the same thing. When a human looks at a stop sign, they recognise it through shape, context, colour, and accumulated visual experience. When an AI classifies the same stop sign, it computes numerical patterns across pixel values. These two perceptual processes exploit different features — and that gap is exactly what input manipulation attacks target.

An adversarial example is an input that has been mathematically crafted to exploit the AI's feature-detection process. The changes are designed to be imperceptible to humans — a slight shift in pixel values, a character substitution, a tiny audio perturbation — while being precisely calculated to flip the model's prediction. The model has not been compromised; it is functioning exactly as trained. The attack exploits the mismatch between human and machine perception.

Fundamental insight: Adversarial examples are not bugs in the model. They reveal a structural property: AI decision boundaries are sharp and can be crossed with minimal, human-imperceptible changes.

2. How Adversarial Examples Are Generated — The Gradient Method

To generate an adversarial example with white-box access (access to model weights and architecture), an attacker computes the gradient of the model's loss function with respect to the input. The gradient describes which direction each input feature (pixel, token, byte) should be changed to most increase the model's error. The attacker then applies a small perturbation in that direction — just enough to cross the decision boundary, not enough for humans to notice.

Common algorithms include the Fast Gradient Sign Method (FGSM), which applies a single gradient step, and Projected Gradient Descent (PGD), which iterates multiple steps to find a stronger adversarial example while keeping the perturbation within a bounded distance from the original. Both methods are well-documented and their implementations are publicly available, making them accessible to adversaries with moderate technical capability.

3. Attack Types Classified by Knowledge and Goal

Attack Type	Attacker's Knowledge	Goal	Security Example
White-box	Full access to model weights, architecture, and training data	Craft optimal adversarial examples with minimal perturbation	Insider generating adversarial malware to evade an internal AI detector
Black-box	Query access only — send inputs, observe output labels or	Infer decision boundaries through repeated probing	External attacker querying a public API to map detection thresholds

	scores		
Transfer attack	Access to a surrogate model; no access to target	Generate adversarial examples on surrogate that also fool target	Attacker uses open-source model to craft evasion samples that work against production system
Evasion	Any	Cause misclassification at inference time	Modifying malware byte sequences to evade AI-based AV
Physical-world	None needed at deployment time	Create physical artefacts that fool AI sensors	Printed adversarial patch on clothing defeating facial recognition at an access gate

4. Real-World Incident — Physical Adversarial Examples

Real-World Incident — Stop Sign Adversarial Patch Study, University of Washington (2017): Researchers demonstrated that printed adversarial patches attached to physical stop signs caused an AI vision classifier to misclassify the stop sign as a speed limit sign with high confidence, from multiple viewing angles and distances. The patches looked like graffiti to human observers. This demonstrated that adversarial attacks are not purely theoretical — they survive the analogue-digital transition and can be deployed in physical environments without any access to the target system's internals. Security AI relying on image or video feeds faces the same attack surface.

5. Evasion in Cybersecurity — Malware and Network Traffic

The most operationally significant input manipulation attacks for SecAI+ candidates involve malware evasion and network traffic manipulation. Malware authors use several obfuscation techniques to defeat AI-based detection:

- **Code packing and encryption:** Wrapping malicious code in a packer compresses and encrypts the payload. AI classifiers trained on raw byte features see an entirely different statistical profile.
- **Polymorphism:** The malware rewrites its own non-functional code on each execution, changing its binary signature while preserving behaviour. AI trained on byte n-grams is confused by each new variant.
- **Metamorphism:** More advanced than polymorphism — the malware rewrites its functional code too, changing structure and instruction sequences while preserving semantics. Behaviour-based AI detectors are more resilient against this than signature-based ones.
- **Adversarial byte injection:** Adding carefully computed bytes to a Portable Executable (PE) file that do not affect execution but shift the file's feature representation away from the malicious class boundary.

6. Transferability — Why It Matters for Defence

Adversarial examples transfer across models. An example crafted against a publicly available surrogate model (e.g., an open-source malware classifier) often successfully fools a private production model with

a different architecture. This transferability means that keeping a production model's architecture and weights secret does not prevent adversarial attacks — attackers can craft evasion samples without ever touching the target system.

Transferability varies with model similarity: attacks transfer more reliably between models trained on similar data with similar architectures. But even cross-architecture transfer rates are high enough to be operationally significant. Defence strategies cannot rely on security-through-obscurity of the model itself.

Implication for defenders: Model secrecy is not a defence against input manipulation. Assume adversaries can generate effective adversarial examples without access to your model by using publicly available surrogates.

7. Black-Box Attack Mechanics — Querying Your Way to Evasion

In a black-box attack, the adversary has no knowledge of model internals but can submit queries and observe responses. By systematically varying inputs and observing how classifications change, the attacker builds a functional map of the model's decision boundary. Two main strategies are used:

- **Decision-based attacks:** The attacker observes only the final classification label (not confidence scores). Attacks like Boundary Attack iteratively move an adversarial input toward the natural input while staying in the misclassified region.
- **Score-based attacks:** If the attacker can observe confidence scores or logits, gradient estimation techniques like NES (Natural Evolution Strategies) approximate the gradient without direct access to weights, enabling gradient-based optimisation with only query access.

Rate limiting and query monitoring are therefore legitimate defensive measures: restricting the number of queries per user or per time window degrades the attacker's ability to map the decision boundary. High query volumes with small input variations should trigger alerts.

8. Defence Mechanisms Compared

Defence	Mechanism	Strengths	Weaknesses
Input validation	Reject inputs outside expected format, size, or statistical distribution	Eliminates malformed and out-of-distribution manipulation attempts	Does not catch subtle adversarial examples within valid distribution
Adversarial training	Include adversarial examples in the training set; train model to classify them correctly	Directly improves model robustness; reduces attack success rate	Computationally expensive; requires ongoing regeneration as attacks evolve
Input preprocessing / smoothing	Apply JPEG compression, noise addition, or feature squeezing before classification	Disrupts carefully computed perturbations	Reduces accuracy on clean inputs; not adaptive to new attack algorithms

Ensemble methods	Aggregate predictions from multiple diverse models	Attacker must fool all models simultaneously; increases attack cost	Diversity must be genuine — correlated ensembles provide little benefit
Anomaly detection on inputs	Flag inputs statistically distant from training distribution	Catches adversarial examples that occupy unusual input space	Not all anomalies are attacks; not all attacks are anomalous
Query rate limiting	Restrict queries per source per time window	Degrades black-box attack feasibility	Does not stop white-box or transfer attacks

9. Behavioural Monitoring — Catching Attackers During Reconnaissance

Adversarial attack preparation has a detectable behavioural signature. Before a successful evasion attack, an attacker probing a black-box system typically submits many queries with systematically varied inputs — small perturbations, encoded variants, or format mutations — to map the decision boundary. This reconnaissance phase generates observable patterns:

- High query volume from a single API key or IP address within a short window.
- Inputs that are structurally similar to each other but differ in small, systematic ways.
- Repeated submissions of near-duplicate files with minor byte-level variations.
- Queries that oscillate between two classification outcomes as the attacker searches for a boundary.

SIEM rules, API gateway analytics, and AI-specific monitoring platforms can flag these patterns. Detecting reconnaissance early — before a fully optimised adversarial example is crafted — allows defenders to rate-limit, block, or investigate before a successful evasion is achieved.

10. Memory Anchor — The Optical Illusion Analogy

Adversarial examples are to AI what optical illusions are to human vision. An optical illusion exploits the assumptions your visual cortex makes about edges, depth, and lighting. You see something that is not there because your brain's shortcuts were designed for common scenarios, not adversarial ones.

Adversarial examples exploit the assumptions baked into the AI's training process. The model's decision boundaries were carved out by gradient descent on a training set that did not include crafted adversaries. The attack finds the gap between what the training process optimised for and what the real world can present.

Key Terms Glossary

Term	Definition
Adversarial example	An input deliberately crafted with small perturbations to cause an AI model to misclassify it
Evasion attack	An input manipulation attack targeting the inference phase, causing the model to misclassify malicious content as benign
White-box attack	An adversarial attack where the attacker has full access to model weights, architecture, and gradients

Black-box attack	An adversarial attack using only query access to the model, inferring decision boundaries from observed outputs
Transfer attack	An adversarial example crafted against a surrogate model that also fools a different target model
FGSM (Fast Gradient Sign Method)	A single-step white-box attack that computes the gradient of the loss with respect to the input and applies a bounded perturbation
Polymorphism	Malware technique where non-functional code is rewritten each execution to avoid signature detection
Adversarial training	Defence technique that includes adversarial examples in the training set to improve model robustness
Ensemble defence	Using multiple diverse models whose aggregated predictions are harder for adversarial examples to simultaneously fool
Input preprocessing	Applying transforms (compression, noise, squeezing) to inputs before classification to disrupt adversarial perturbations

Quick Revision Checklist

- Adversarial examples exploit the gap between human perception and AI feature computation — not a bug, but a structural property of how models learn.
- White-box attacks use gradient information; black-box attacks use query probing; transfer attacks use surrogate models.
- Evasion attacks target malware detection, spam filters, and intrusion detection — any AI classifier in a security role.
- Adversarial examples transfer across models, so model secrecy is not a reliable defence.
- Physical-world adversarial attacks work against image and audio AI systems — camera feeds, voice authentication, access control.
- Defences include: input validation, adversarial training, input preprocessing, ensemble methods, anomaly detection, and query rate limiting.
- Each defence has weaknesses — no single defence is sufficient; layered defence-in-depth is required.
- Behavioural monitoring during the reconnaissance phase can detect black-box attack preparation before evasion succeeds.
- Polymorphism, metamorphism, packing, and adversarial byte injection are the primary malware evasion techniques against AI detectors.

10 Exam-Based MCQs — Input Manipulation Attacks

These questions are written in real exam style — scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. *Scenario:* A security analyst reviewing API gateway logs for an ML-based network intrusion detection system notices that a single external IP address submitted 4,200 queries over six hours. The queries are structurally similar network flow records that differ only in small, systematic variations of port numbers

and packet sizes. Classification outputs alternate between 'benign' and 'suspicious' in a pattern that suggests boundary exploration. What type of attack is MOST likely being carried out, and what defensive measure would BEST disrupt it?

- A) A white-box evasion attack; implement model weight encryption to prevent gradient computation
- B) A black-box boundary-mapping attack; implement query rate limiting and alert on high-volume systematic probing
- C) A training data poisoning attack; audit the training dataset for mislabelled samples
- D) A transfer attack; replace the public API with a private endpoint accessible only to internal systems

Answer: B **Explanation:** B is correct because the described behaviour — high query volume with small, systematic input variations — is the characteristic signature of a black-box attack mapping the model's decision boundary. Rate limiting degrades the attacker's ability to collect enough observations to optimise an adversarial example, and alerting enables investigation. A is incorrect because white-box attacks use gradient access and do not require repeated queries; model weight encryption does not prevent query-based probing. C is incorrect because training data poisoning occurs during the training phase, not during inference; query probing is an inference-phase behaviour. D is incorrect because making an endpoint private does not prevent transfer attacks and may not prevent a determined adversary from gaining access through other means.

Q2. Scenario: A malware analyst discovers that a threat actor has been submitting PE files to an organisation's AI-based malware sandbox that are classified as benign despite containing a known RAT payload. Static analysis confirms the files contain the payload; the AI classifier consistently rates them as clean. The analyst determines the actor has added precisely computed byte sequences to non-executable sections of the PE file, shifting the file's feature vector away from the malicious class boundary. Which defensive measure would MOST directly address the evasion technique being used?

- A) Update the signature database to include the new file hash
- B) Implement adversarial training by incorporating adversarially perturbed PE files in the malware classifier's training set
- C) Block the originating IP address from submitting files to the sandbox
- D) Apply JPEG compression to all submitted files before analysis to disrupt pixel-level changes

Answer: B **Explanation:** B is correct because adversarial training — including adversarially perturbed samples in the training set — directly improves the model's robustness against the class of perturbations being used. The model learns that inputs near the malicious class boundary should still be classified as malicious. A is incorrect because updating the signature hash addresses signature-based detection, not the AI classifier's feature-based classification; the perturbed file has a different hash but the AI evasion is independent of the hash. C is incorrect because blocking the IP is an access control measure, not a defence against the evasion technique itself; adversaries change IP addresses. D is incorrect because JPEG compression is an image preprocessing defence; it does not apply to PE file byte sequences.

Q3. Scenario: A red team crafted adversarial malware samples using a publicly available open-source malware classifier as a surrogate. When submitted to the organisation's private, internally developed malware detection model — which uses a different architecture and was trained on proprietary data — 68% of the samples were misclassified as benign. The organisation had not publicly disclosed any details about their model. Which of the following BEST explains why the adversarial samples work against the organisation's private model even though they were crafted against the public model?

- A) The organisation's model was trained on the same open-source dataset as the public model, making their weights identical
- B) Adversarial examples transfer across models because different models trained on similar data learn similar decision boundary structures
- C) The attacker gained access to the organisation's model weights through an earlier breach and used them for gradient computation
- D) The samples work because the public model has higher accuracy than the organisation's model, meaning its adversarial examples are universally effective

Answer: B Explanation: B is correct because adversarial transferability arises from the phenomenon that models trained on similar data tend to learn similar feature representations and decision boundaries. Perturbations that push inputs across one model's boundary often cross similar boundaries in other models. A is incorrect because models trained on similar datasets do not have identical weights; weight identity is not required for transfer. C is incorrect because the scenario specifies the organisation's model weights are not publicly accessible; attributing the transfer to a prior breach introduces an unsupported assumption. D is incorrect because relative accuracy does not determine transferability; the mechanism is shared decision boundary geometry, not accuracy ranking.

Q4. Which statement BEST describes why adversarial examples are considered a fundamental challenge rather than a solvable bug?

- A) Adversarial examples can only be created by nation-state attackers with access to advanced hardware
- B) Current AI architectures optimise for accuracy on training distributions; adversarially crafted inputs exploit the sharp, exploitable decision boundaries that result from this optimisation
- C) Adversarial examples are caused by insufficient training data; adding more training samples will eventually eliminate them
- D) The problem is fundamental only for image classifiers and does not affect text or binary file classifiers

Answer: B Explanation: B is correct because gradient descent optimisation produces models with sharp, precise decision boundaries optimised for the training distribution. These boundaries are inherently exploitable by inputs engineered to cross them — the sharpness that enables high accuracy also enables adversarial perturbation. A is incorrect because adversarial example generation tools are publicly available and require only modest hardware. C is incorrect because adding training data reduces hallucination and improves generalisation but does not eliminate adversarial vulnerability; the mathematical structure of decision boundaries remains exploitable. D is incorrect because adversarial examples have been demonstrated across image, text, audio, and binary classification domains.

Q5. An attacker attaches a printed pattern to clothing worn while walking through an area monitored by AI-powered facial recognition. The system fails to identify the individual. Which type of attack does this represent?

- A) A model poisoning attack targeting the facial recognition system's training pipeline
- B) A physical-world adversarial example exploiting the gap between human and AI perception
- C) A black-box query attack mapping the facial recognition system's classification thresholds
- D) A denial-of-service attack preventing the facial recognition system from processing images

Answer: B **Explanation:** B is correct because the printed pattern is a physical adversarial example — a crafted perturbation that appears as ordinary clothing to humans while causing the AI vision system to fail to detect or correctly classify the face. This category of attack has been demonstrated in research and represents a real operational threat to physical security systems relying on AI cameras. A is incorrect because model poisoning targets the training phase; no access to training data or pipelines is involved here. C is incorrect because the attacker is not submitting queries to a model API; the attack occurs passively in physical space. D is incorrect because the system is not prevented from processing images — it processes them and produces incorrect classifications.

Q6. Which property of adversarial examples makes model secrecy an insufficient defence against input manipulation attacks?

- A) Adversarial examples can be generated by guessing model architecture from the vendor's marketing materials
- B) Adversarial examples transfer across models, enabling attackers using surrogate models to generate effective attacks without accessing the target
- C) All commercial AI models use the same underlying architecture, making adversarial examples universally applicable
- D) Regulatory requirements mandate that model architectures be disclosed in security product documentation

Answer: B **Explanation:** B is correct because transferability means an adversarial example crafted against a publicly accessible surrogate model often fools a private target model. The attacker never needs access to the target's weights or architecture. A is incorrect because reverse-engineering architecture from marketing materials is speculative and not the mechanism by which transfer attacks succeed. C is incorrect because commercial AI models use diverse architectures; universality of architecture is not the transfer mechanism. D is incorrect because there is no such universal disclosure requirement for commercial security products.

Q7. Which of the following input preprocessing defences is MOST likely to disrupt adversarial perturbations in image-based security AI?

- A) Encrypting images before analysis to prevent attackers from observing classification outputs
- B) Applying JPEG compression or feature squeezing before classification to smooth out precisely computed pixel perturbations

- C) Converting all colour images to greyscale to reduce the number of features available for adversarial manipulation
- D) Increasing the resolution of input images to ensure the model has maximum information for classification

Answer: B **Explanation:** B is correct because JPEG compression and feature squeezing remove high-frequency details — exactly the type of precise, small-magnitude changes that adversarial perturbations rely on. The adversarial example depends on exact pixel values; lossy compression disrupts those values and can neutralise the attack. A is incorrect because encrypting images before analysis would prevent the model from classifying them at all; this is not a practical defence. C is incorrect because greyscale conversion reduces information but does not specifically target adversarial perturbation patterns and may reduce model accuracy on legitimate tasks. D is incorrect because increasing resolution provides more information but also increases the adversarial attack surface.

Q8. What is the PRIMARY advantage of using an ensemble of diverse models as a defence against adversarial examples?

- A) Diverse models have lower accuracy individually, so attackers targeting them gain less useful information
- B) An adversarial example must fool all models in the ensemble simultaneously, which requires a more complex attack and is harder to achieve
- C) Ensemble models share weights, so patching one model automatically patches all others in the ensemble
- D) Ensemble methods eliminate the need for adversarial training because diversity replaces the need for robust individual classifiers

Answer: B **Explanation:** B is correct because each model in a diverse ensemble has different decision boundaries. An adversarial example optimised against one model's boundary is unlikely to simultaneously cross the boundaries of other models trained with different architectures or data. The attacker's optimisation problem becomes significantly harder. A is incorrect because the advantage of ensembles is the diversity of decision boundaries, not reduced individual accuracy. C is incorrect because ensemble models are independent and do not share weights; patching one does not affect the others. D is incorrect because ensemble diversity and adversarial training address complementary aspects of robustness; neither makes the other unnecessary.

Q9. Which malware obfuscation technique is MOST resistant to AI classifiers that rely on behavioural analysis rather than byte-pattern features?

- A) Code packing, which compresses and encrypts the malware payload
- B) Metamorphism, which rewrites the malware's functional code while preserving its malicious behaviour
- C) Polymorphism, which rewrites non-functional code sections to change the binary signature
- D) Adversarial byte injection, which adds computed bytes to non-executable PE file sections

Answer: B **Explanation:** B is correct because metamorphism rewrites the functional code — the actual

instruction sequences that implement malicious behaviour — while preserving the semantic outcome. Behavioural AI classifiers that look at code structure, call graphs, or instruction patterns are more challenged by metamorphism than by techniques that only change surface-level features. A is incorrect because packing changes the static byte profile but the unpacked behaviour is typically similar; behavioural analysis on the unpacked execution can still detect the malware. C is incorrect because polymorphism only rewrites non-functional code, leaving functional behaviour largely intact and detectable by behavioural analysis. D is incorrect because adversarial byte injection targets byte-feature classifiers in the static analysis phase; it does not affect behavioural classification.

Q10. Scenario: *A large financial institution's AI-based spam filter has a 97% detection rate for phishing emails. Over two weeks, a new phishing campaign achieves an 81% bypass rate. Analysis reveals the campaign emails contain zero-width Unicode characters interspersed through key words and deliberate leetspeak substitutions. Human readers interpret the text correctly, but the feature extraction pipeline produces vectors that score as legitimate email.* An AI-based spam filter begins misclassifying a phishing campaign as legitimate email. Investigation reveals the emails contain invisible Unicode characters and deliberate misspellings that humans interpret correctly but that shift the text's feature vector. Which TWO actions BEST address this attack? (Choose two.)

- A) Apply text normalisation and Unicode sanitisation as a preprocessing step before classification
- B) Retrain the classifier on adversarially crafted email examples including similar obfuscation patterns
- C) Replace the AI-based spam filter with a rule-based system using keyword matching
- D) Increase the classifier's confidence threshold so it only flags emails it is very certain are phishing

Answer: A Explanation: A and B are correct. A addresses the immediate problem: normalising Unicode characters and sanitising invisible characters removes the obfuscation before it reaches the classifier, disrupting the feature-vector manipulation. B addresses the underlying robustness problem: adversarial training with similar obfuscation patterns teaches the model to classify obfuscated phishing correctly. C is incorrect because rule-based keyword matching is more easily evaded than a well-trained classifier and reduces detection of novel phishing patterns. D is incorrect because raising the confidence threshold increases false negatives (missed phishing) without improving robustness against the evasion technique — the adversarially crafted emails will still score below the higher threshold.